10. *On Fundamental Systems of Symmetric Functions—II.*

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(Read before the Physico-Mathematical Society of Japan, April 2, 1927.)

I

For n independent variables

$$x_1,\ x_2,\ \ldots\ldots,\ x_n \tag{1}$$

we denote, as usual, the power sum $x_1^k+x_2^k\ \ldots\ldots\ +x_n^k$, k being a positive integer, by S_k.

If a system of n power sums

$$S_{\nu_1},\ S_{\nu_2},\ \ldots\ldots,\ S_{\nu_n},\ (\nu_1<\nu_2<\ldots\ldots<\nu_n) \tag{2}$$

has the nature that any rational symmetric function of (1) can be expressed by a rational function of (2), then the system (2) is called a *fundamental system* (of power sums).

Various fundamental systems have been given by several authors, and I have also obtained a kind of fundamental systems(1), which includes all the systems before obtained.

Here I have to give the same result in a simpler manner and to show that the systems thus obtained include all the possible systems in a certain restricted sense.

II

If the system (2) is fundamental, then it is necessary that each of n elementary symmetric functions

$$p_i=\sum x_1x_2\ldots\ldots x_i \qquad i=1,\ 2,\ldots\ldots,\ n \tag{3}$$

can be expressed rationally by (2) in the form

$$p_i=\frac{A_i(S_{\nu_1},\ S_{\nu_2},\ \ldots,\ S_{\nu_n},)}{B_i(S_{\nu_1},\ S_{\nu_2},\ \ldots,\ S_{\nu_n})}, \tag{4}$$

where A_i and B_i are polynomials of (2). Hence if the values of S_{ν_i}'s of (2) are given, the values of p_i's of (3) must, in general, be determined uniquely. The exception may occur only when some of B_i's of (4) should vanish, in which case $x_1,\ x_2,\ldots\ldots,\ x_n$ and consequently $p_1,\ p_2,\ldots\ldots,\ p_n$ have

(1) This Journal, 2 (1925), 69.

some particular relations among themselves. And the unique determination of p_i's is equivalent to the unique determination of the functional form

$$f(x)=1+p_1x+p_2x^2+\cdots\cdots+p_nx^n$$
$$=(1+x_1x)(1+x_2x)\cdots\cdots(1+x_nx). \tag{5}$$

Conversely, if the form of (5) is determined uniquely, in general, from (2), then (2) must be a fundamental system. For each of (3) is evidently an algebraic function of (2) ([2]), and it is uniform, so that it must be a rational function of (2). And any other rational symmetric function of (1) is known to be a rational function of (3).

Now it is easily seen that

$$\log f(x)=\sum \log(1+x_ix)$$
$$=S_1x-\frac{1}{2}S_2x^2+\frac{1}{3}S_3x^3-\cdots\cdots+(-1)^{m-1}\frac{1}{m}S_mx^m\cdots\cdots \tag{6}$$

Consequently, if $f(x)$ is determined from (2), then $\log g(x)$, where $g(x)$ is a polyniomial of the form

$$g(x)=1+q_1x+q_2x^2+\cdots\cdots+q_nx^n,$$

has the expansion of the similar form as (6) in which the coefficients of x^{ν_1}, x^{ν_2}, ..., x^{ν_n} are identical with those of (6), when and only when $f(x)$ is identical with $g(x)$. Namely, if in the expansion

$$\log f(x)-\log g(x)=\log\frac{f(x)}{g(x)}$$
$$=t_1x+t_2x^2+\cdots\cdots+t_mx^m+\cdots\cdots \tag{7}$$

t_{ν_1}, t_{ν_2}, ..., t_{ν_n} are equal to zero, we must have $f(x)\equiv g(x)$.

The exeption may occur only when some particular relations among the coefficients of $f(x)$(it is same to say of $g(x)$) exist. Thus we get

Lemma. The n power sums S_{ν_1}, S_{ν_2},....., S_{ν_n} *form a fundamental*

([2]) The functions (2) are independent with respect to the variables (1). For the corresponding Jacobian is

$$\begin{vmatrix} \nu_1x_1^{\nu_1-1}\cdots\cdots\nu_1x_n^{\nu_1-1} \\ \cdots\cdots\cdots\cdots\cdots\cdots \\ \nu_nx_1^{\nu_n-1}\cdots\nu_nx_n^{\nu_n-1} \end{vmatrix}=\frac{\nu_1\cdots\cdots\nu_n}{x_1\cdots\cdots x_n}\begin{vmatrix} x_1^{\nu_1}\cdots\cdots x_n^{\nu_1} \\ \cdots\cdots\cdots\cdots \\ x_1^{\nu_n}\cdots\cdots x_n^{\nu_n} \end{vmatrix}$$

And this is not identically zero since the determinant of the right hand side has the term $x_1^{\nu_1}x_2^{\nu_2}\cdots x_n^{\nu_n}$ whose coefficient is 1.

system when and only when the identity

$$log\frac{1+p_1x+p_2x^2+\cdots\cdots+p_nx^n}{1+q_1x+q_2x^2+\cdots\cdots+q_nx^n}=t_1x+t_2x^2+\cdots\cdots+t_mx^m+\cdots\cdots,$$

where $t_{\nu_1}=t_{\nu_2}=\cdots\cdots=t_{\nu_n}=0,$

gives the result that either

$$p_1=q_1,\ p_2=q_2,\ \cdots\cdots,\ p_n=q_n$$

or some particular relations among $p_1, p_2, \cdots\cdots, p_n$ *or among* $q_1, q_2, \cdots\cdots, q_n$ *exist*([3]).

III

Let a sequence which consists of zero and some of positive integers, having the nature that the sum of any two (so also more than two) integers (they may be equal) of the sequence also belongs to the sequence, be called an *additive sequence.*

If $(0, \mu_1, \mu_2, \cdots\cdots)$ is an additive sequence, then *the sum or product of two or more power series of the form*

$$c_0+c_1x^{\mu_1}+c_2x^{\mu_2}+\cdots\cdots+c_mx^{\mu_m}+\cdots\cdots,$$

namely the series whose indices form the additive sequence, is also a power series of the same form. For the index of a term of the sum evidently belongs to the same sequence and the index of a term of the product is evidently the sum of the indices of some terms of the factors, so that it belongs also to the same sequence.

We are then able to prove

Theorem I. If a sequence consisting of all non-negative integers, different from the given n positive integers $\nu_1, \nu_2, \cdots\cdots, \nu_n$, *is additive, then* $S_{\nu_1}, S_{\nu_2}, \cdots\cdots, S_{\nu_n}$ *form a fundamental system.*

Suppose that the sequence $M(0, \mu_1, \mu_2, \cdots\cdots)$ of all non-negative integers, different from $\nu_1, \nu_2, \cdots\cdots, \nu_n$, be additive. And suppose that in the expansion

$$log\frac{1+p_1x+p_2x^2+\cdots\cdots+p_nx^n}{1+q_1x+q_2x^2+\cdots\cdots+q_nx^n}=t_1x+t_2x^2+\cdots\cdots+t_mx^m+\cdots\cdots, \qquad (8)$$

the coefficients $t_{\nu_1}, t_{\nu_2}, \cdots\cdots, t_{\nu_n}$ be all zero.

Then the indices of the power series of (8) belong to the sequence M. Consequently, in the relation

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([3]) The proof of this paragraph is identical with that of my former paper.

$$\frac{1+p_1x+p_2x^2+\cdots\cdots+p_nx^n}{1+q_1x+q_2x^2+\cdots\cdots+q_nx^n}=e^{t_1x+t_2x^2+\cdots\cdots}$$

$$=1+(t_1x+t_2x^2+\cdots\cdots)+\frac{1}{\lfloor 2}(t_1x+t_2x^2+\cdots\cdots)^2+\cdots\cdots$$

$$=1+\tau_1x+\tau_2x^2+\cdots\cdots+\tau_mx^m+\cdots\cdots, \tag{9}$$

the indices of the power series (9) must also belong to the sequence M. For it is composed of the series of the same kind (the series (8)) by means of the additions and of the multiplications.

Now the rational function (9) can be written in the form

$$\frac{1+p_1x+p_2x^3+\cdots\cdots+p_nx^n}{1+q_1x+q_2x^2+\cdots\cdots+q_nx^n}=1-\left(\frac{A_1y_1x}{1+y_1x}+\frac{A_2y_2x}{1+y_2x}+\cdots\cdots+\frac{A_ny_nx}{1+y_nx}\right)$$

$$=1-(A_1y_1+A_2y_2+\cdots\cdots+A_ny_n)x$$

$$+(A_1y_1^2+A_2y_2^2+\cdots\cdots+A_ny_n^2)x^2-\cdots\cdots \tag{10}$$

where $$1+q_1x+q_2x^2+\cdots\cdots+q_nx^n=(1+y_1x)(1+y_2x)\cdots\cdots(1+y_nx^n),$$

and $A_1, A_2, \cdots\cdots, A_n$ are some constants.

Equating the coefficients of $x^{\nu_1}, x^{\nu_2}, \cdots\cdots, x^{\nu_n}$ in the two power series (9) and (10), we get

$$A_1y_1^{\nu_i}+A_2y_2^{\nu_i}+\cdots\cdots+A_ny_n^{\nu_i}=0, \quad i=1, 2, \cdots\cdots, n \tag{11}$$

Now the determinant

$$\Delta=\begin{vmatrix} y_1^{\nu_1} & y_2^{\nu_1} & \cdots\cdots & y_n^{\nu_1} \\ y_1^{\nu_2} & y_2^{\nu_2} & \cdots\cdots & y_n^{\nu_2} \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & \cdots\cdots \\ y_1^{\nu_n} & y_2^{\nu_n} & \cdots\cdots & y_n^{\nu_n} \end{vmatrix}$$

is not identically zero(4). Hence from (11) we get either the result

$$A_i=0, \quad i=1, 2, \cdots\cdots, n$$

namely $$p_i=q_i, \quad i=1, 2, \cdots\cdots, n$$

or the relation $\Delta=0$ among $y_1, y_2, \cdots\cdots, y_n$, which is the relation among $q_1, q_2, \cdots\cdots, q_n$.

Thus by the lemma we see that $S_{\nu_1}, S_{\nu_2}, \cdots\cdots, S_{\nu_n}$ form a fundamental system.

IV

If a system of n power sums $S_{\nu_1}, S_{\nu_2}, \cdots\cdots, S_{\nu_n}$ forms a fundamental

(4) See the preceding foot note.

system and moreover a system of m power sums S_{ν_1}', S_{ν_2}', ..., S'_{ν_m} with respect to m variables $x_1, x_2, \ldots\ldots, x_m$ forms also a fundamental system for all values of m less than n, then we call $S_{\nu_1}, S_{\nu_2}, \ldots\ldots, S_{\nu_n}$ the *normal fundamental systsm.*

Now we can prove

Theorem **2.** *In order that a system of n power sums* $S_{\nu_1}, S_{\nu_2}, \ldots, S_{\nu_n}$ *forms a normal fundamental system, it is necessary and sufficient that the sequence consisting of all the non-negative integers, different from* $\nu_1, \nu_2, \ldots\ldots, \nu_n$, *is additive.*

That the condition is sufficient is evident, for the sequence obtained from the additive sequence $(0, \mu_1, \mu_2, \ldots\ldots)$ of the preceding paragraph by adding n-m numbers $\nu_{m+1}, \nu_{m+2}, \ldots\ldots, \nu_n$ is also additive, so that the m power sums $S'_{\nu_1}, S'_{\nu_2}, \ldots\ldots, S'_{\nu_m}$ form a fundamental system with respect to m variables.

To prove the necessity of the theorem we make use of the mathematical induction.

The condition is necessary in the case $n=1$, for only the possible fundamental system is $S_1 \equiv x_1$, and $(2, 3, \ldots\ldots)$ is an additive sequence.

Assume that the condition is necessary for the case of $n-1$ and that the n power sums $S_{\nu_1}, S_{\nu_2}, \ldots\ldots, S_{\nu_n}$ form a normal fundamental system.

Since $S'_{\nu_1}, S'_{\nu_2}, \ldots\ldots, S'_{\nu_{n-1}}$ form a normal fundamental system, the sequence M' consiting of all the non-negative integers different from $\nu_1, \nu_2, \ldots\ldots, \nu_{n-1}$, namely the sequence obtained from $M(0, \mu_1, \mu_2, \ldots\ldots)$ by adding an integer ν_n is additive.

Now suppose that the sequence M is not additive, then ν_n must be equal to the sum of two or more of $\mu_1, \mu_2, \ldots\ldots$. Let, for example,

$$\nu_n = \mu_a + \mu_b + \mu_c + \mu_d.$$

Since $\mu_a + \mu_b$ and $\mu_c + \mu_d$ also belong to M' and less than ν_n, each of them must be equal respectively to some of $\mu_1, \mu_2, \ldots\ldots$, say μ_α, μ_β, so we get the relation

$$\nu_n = \mu_\alpha + \mu_\beta,$$

whose right hand side is the sum of the two of $\mu_1, \mu_2, \ldots\ldots$.

Without loss of generality, we assume that there are just two such relations namely

$$\nu_n = \mu_\alpha + \mu_\beta \text{ and } \nu_n = \mu_\gamma + \mu_\delta. \tag{12}$$

Now consider again the rational function

$$\frac{1+p_1x+p_2x^2+\ldots\ldots+p_nx^n}{1+q_1x+q_2x^2+\ldots\ldots+q_nx^n}=1-\left(\frac{A_1y_1x}{1+y_1x}+\frac{A_2y_2x}{1+y_2x}+\ldots+\frac{A_ny_nx}{1+y_nx}\right)$$

$$=1-(A_1y_1+A_2y_2+\ldots\ldots+A_ny_n)x+(A_1y_1^2+A_2y_2^2+\ldots\ldots+A_ny_n^2)x^2-\ldots\ldots$$

$$=1+\tau_1x+\tau_2x^2+\ldots\ldots+\tau_mx^m+\ldots\ldots \tag{13}$$

where $1+q_1x+q_2x^2+\ldots\ldots+q_nx^n=(1+y_1x)(1+y_2x)\ldots\ldots(1+y_nx)$.

Take $A_1, A_2, \ldots\ldots, A_n$ and consequently $p_1, p_2, \ldots\ldots, p_n$ so as to satisfy the equations

$$(\overset{\bullet}{-}1)^{\nu_i}\tau_{\nu_i}=\sum_K A_k y_k^{\nu_i}=0, \qquad i=1, 2, \ldots\ldots, n-1 \tag{14}$$

which determine the ratio $A_1 : A_2 : \ldots\ldots : A_n$. We get

$$A_k=D_k\lambda, \qquad k=1, 2, \ldots\ldots, n \tag{15}$$

where λ is a variable parameter and

$$D_k=(-1)^k\begin{vmatrix} y_1^{\nu_1} \ldots\ldots & y_{k-1}^{\nu_1} y_{k+1}^{\nu_1} \ldots\ldots y_n^{\nu_1} \\ \ldots\ldots\ldots\ldots & \\ y_1^{\nu_{n-1}} \ldots y_{k-1}^{\nu_{n-1}} & y_{k+1}^{\nu_{n-1}} \ldots y_n^{\nu_{n-1}} \end{vmatrix} \tag{16}$$

and hence

$$\tau_i=(-1)^i\sum_K A_k y_k^i=(-1)^{n+i}\begin{vmatrix} y_1^{\nu_1} y_2^{\nu_1} \ldots\ldots\ldots y_n^{\nu_1} \\ \ldots\ldots\ldots\ldots \\ y_1^{\nu_{n-1}} y_2^{\nu_{n-1}} \ldots y_n^{\nu_{n-1}} \\ y_1^i \; y_2^i \ldots\ldots\ldots\ldots y_n^i \end{vmatrix}\lambda$$

$$=(-1)^{n+i}\Delta_i\lambda. \tag{17}$$

And we have

$$\log\frac{1+p_1x+p_2x^2+\ldots+p_nx^n}{1+q_1x+q_2x^2+\ldots+q_nx^n}=(\tau_1x+\tau_2x^2+\ldots)-\frac{1}{2}(\tau_1x+\tau_2x^2+\ldots)^2$$

$$+\frac{1}{3}(\tau_1x+\tau_2x^2+\ldots)^3-\ldots\ldots \tag{18}$$

In the power series

$$\tau_1x+\tau_2x^2+\ldots\ldots$$

the coefficients of $x^{\nu_1}, x^{\nu_2}, \ldots, x^{\nu_{n-1}}$ are zero and hence the power series consists of the terms whose indices are the numbers of the additive sequence M'. Therefore the same is true for the expansion of (18).

The coefficient of x^{ν_n} in the expansion of (18) is

$$\tau_{\nu_n}-(\tau_{\mu_\alpha}\tau_{\mu_\beta}+\tau_{\mu_\gamma}\tau_{\mu_\delta})+\cdots\cdots\cdots\cdots(^5)$$
$$=(-1)^{n+\nu_n}\{\Delta_{\nu_n}\lambda-(\Delta\mu_\alpha\Delta\mu_\beta+\Delta\mu_\gamma\Delta\mu_\delta)\lambda^2+\cdots\cdots\}, \qquad (19)$$

where the dotted part contains only the powers of λ higher than 2.

First we see evidently that

$$\Delta\nu_n=\begin{vmatrix} y_1^{\nu_1}y_2^{\nu_1}\cdots\cdots\cdots y_n^{\nu_1} \\ \cdots\cdots\cdots\cdots\cdots\cdots \\ y_1^{\nu_{n-1}}y_2^{\nu_{n-1}}\cdots y_n^{\nu_{n-1}} \\ y_1^{\nu_n}y_2^{\nu_n}\cdots\cdots\cdots y_n^{\nu_n} \end{vmatrix} \qquad (20)$$

is not identically zero.

And next

$$\Delta\mu_\alpha\Delta\mu_\beta+\Delta\mu_\gamma\Delta\mu_\delta$$
$$=\begin{vmatrix} y_1^{\nu_1}\cdots\cdots y_n^{\nu_1} \\ \cdots\cdots\cdots\cdots \\ y_1^{\nu_{n-1}}\cdots y_n^{\nu_{n-1}} \\ y_1^{\mu_\alpha}\cdots\cdot y_n^{\mu_\alpha} \end{vmatrix}\cdot\begin{vmatrix} y_1^{\nu_1}\cdots\cdots y_n^{\nu_1} \\ \cdots\cdots\cdots\cdots \\ y_1^{\nu_{n-1}}\cdots y_n^{\nu_{n-1}} \\ y_1^{\mu_\beta}\cdots\cdot y_n^{\mu_\beta} \end{vmatrix}+\begin{vmatrix} y_1^{\nu_1}\cdots\cdots y_n^{\nu_1} \\ \cdots\cdots\cdots\cdots \\ y_1^{\nu_{n-1}}\cdots y_n^{\nu_{n-1}} \\ y_1^{\mu_\gamma}\cdots\cdots y_n^{\mu_\gamma} \end{vmatrix}\cdot\begin{vmatrix} y_1^{\nu_1}\cdots\cdots y_n^{\nu_1} \\ \cdots\cdots\cdots\cdots \\ y_1^{\nu_{n-1}}\cdots y_n^{\nu_{n-1}} \\ y_1^{\mu_\delta}\cdots\cdot y_n^{\mu_\delta} \end{vmatrix} \qquad (21)$$

If ν_1 is less than each of μ_α, μ_β, μ_γ, μ_δ, then the smallest power of y_1 in (21) is $y_1^{2\nu_1}$ and its coefficient is

$$\begin{vmatrix} y_2^{\nu_2}\cdots\cdots y_n^{\nu_2} \\ \cdots\cdots\cdots\cdots \\ y_2^{\mu_\alpha}\cdots\cdot y_n^{\mu_\alpha} \end{vmatrix}\cdot\begin{vmatrix} y_2^{\nu_2}\cdots\cdots y_n^{\nu_2} \\ \cdots\cdots\cdots\cdots \\ y_2^{\mu_\beta}\cdots\cdot y_n^{\mu_\beta} \end{vmatrix}+\begin{vmatrix} y_2^{\nu_2}\cdots\cdots y_n^{\nu_2} \\ \cdots\cdots\cdots\cdots \\ y_2^{\mu_\gamma}\cdots\cdot y_n^{\mu_\gamma} \end{vmatrix}\cdot\begin{vmatrix} y_2^{\nu_2}\cdots\cdots y_n^{\nu_2} \\ \cdots\cdots\cdots\cdots \\ y_2^{\mu_\delta}\cdots\cdot y_n^{\mu_\delta} \end{vmatrix}$$

which has the same form as (21). Proceeding in this way, as far as possible, we finally see that either the coefficient of $y_1^{2\nu_1}y_2^{2\nu_2}\cdots y_{\kappa-1}^{2\nu_{\kappa-1}}$ $(\kappa<n-1)$ in (21) is

$$\begin{vmatrix} y_k^{\nu_\kappa}\cdots\cdot y_n^{\nu_\kappa} \\ \cdots\cdots\cdots\cdots \\ y_k^{\mu_\alpha}\cdots\cdot y_n^{\mu_\alpha} \end{vmatrix}\cdot\begin{vmatrix} y_k^{\nu_\kappa}\cdots\cdot y_n^{\nu_\kappa} \\ \cdots\cdots\cdots\cdots \\ y_k^{\mu_\beta}\cdots\cdot y_n^{\mu_\beta} \end{vmatrix}+\begin{vmatrix} y_k^{\nu_\kappa}\cdots\cdot y_n^{\nu_\kappa} \\ \cdots\cdots\cdots\cdots \\ y_k^{\mu_\gamma}\cdots\cdot y_n^{\mu_\gamma} \end{vmatrix}\cdot\begin{vmatrix} y_k^{\nu_\kappa}\cdots\cdots y_n^{\nu_\kappa} \\ \cdots\cdots\cdots\cdots \\ y_k^{\mu_\delta}\cdots\cdot y_n^{\mu_\delta} \end{vmatrix} \qquad (22)$$

where ν_κ is greater than at least one of μ_α, μ_β, μ_γ, μ_δ, or the coefficient of $y_1^{2\nu_1}y_2^{2\nu_2}\cdots y_{n-1}^{2\nu_{n-1}}$ in (21) is

$$|y_n^{\mu_\alpha}|\cdot|y_n^{\mu_\beta}|+|y_n^{\mu_\gamma}|\cdot|y_n^{\mu_\delta}|=2y_n^{\nu_n}$$

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(5) We assume that $\mu_\alpha\neq\mu_\beta$, $\mu_\gamma\neq\mu_\delta$. If for example $\mu_\alpha=\mu_\beta$ we are only to take $\frac{1}{2}(\tau_{\mu}\tau_{\mu_\beta}+2\tau_{\mu_\gamma}\tau_{\mu})$ instead of $\tau_{\mu_\alpha}\tau_{\mu_\beta}+\tau_{\mu_\gamma}\tau_{\mu_\delta}$.

In the later case, we see at once that (21) is not identically zero. So we now consider the former case.

Without loss of generality, we may assume that $\mu_\alpha<\mu_\gamma<\mu_\delta<\mu_\beta$. And then take the smallest integer from each of the following four groups of integers

(i) $\nu_k,\ \nu_{k+1},\ \cdots\cdots,\ \nu_{n-1},\ \mu_\alpha$

(ii) $\nu_k,\ \nu_{k+1},\ \cdots\cdots,\ \nu_{n-1},\ \mu_\beta$

(iii) $\nu_k,\ \nu_{k+1},\ \cdots\cdots,\ \nu_{n-1},\ \mu_\gamma$

(iv) $\nu_k,\ \nu_{k+1},\ \cdots\cdots,\ \nu_{n-1},\ \mu_\delta$

There are four different cases. Namely the smallest numbers are

in (i)	in (ii)	in (iii)	in (iv)
μ_α	ν_κ	ν_κ	ν_κ
μ_α	ν_κ	μ_γ	ν_κ
μ_α	ν_κ	μ_γ	μ_δ
μ_α	μ_β	μ_γ	μ_δ

respectively, according as ν_κ lies in the interval $(\mu_\alpha,\ \mu_\gamma)$, $(\mu_\gamma,\ \mu_\delta)$, $(\mu_\delta,\ \mu_\beta)$ or $(\mu_\beta,\ \infty)$.

The sum of the first and second integers in a row of the above table can not be equal to the sum of the third and fourth integers of the same row, except the fourth. For from each of the three equations

$$\mu_\alpha+\nu_\kappa=\nu_\kappa+\nu_\kappa$$
$$\mu_\alpha+\nu_\kappa=\mu_\gamma+\nu_\kappa$$
$$\mu_\alpha+\nu_\kappa=\mu_\gamma+\mu_\delta$$

we get the result

$$\mu_\alpha=\nu_\kappa$$
$$\mu_\alpha=\mu_\gamma$$
$$\nu_\kappa=\mu_\beta$$

respectively, which is an absurd.

If the first case will occur, $\mu_\alpha+\nu_\kappa$ will be less than $\nu_\kappa+\nu_\kappa$. And the coefficient of the smallest power of y_κ namely of $y_\kappa^{\mu_\alpha+\nu_\kappa}$ in (22) is

$$(-1)^{n-\kappa}\begin{vmatrix} y_{\kappa+1}^{\nu_\kappa} \cdots y_n^{\nu_\kappa} \\ \cdots\cdots\cdots\cdots \\ y_{\kappa+1}^{\nu_{n-1}} \cdots y_{\kappa+1}^{\nu_{n-1}} \end{vmatrix} \cdot \begin{vmatrix} y_{\kappa+1}^{\nu_{\kappa+1}} \cdots y_n^{\nu_{\kappa+1}} \\ \cdots\cdots\cdots\cdots \\ y_{\kappa+1}^{\mu_\beta} \cdots\cdots y_n^{\mu_3} \end{vmatrix}$$

which is not identically zero. Similar result can be seen for the second and third cases.

If the fourth case occurs, $\mu_\alpha+\mu_\beta$ and $\mu_\gamma+\mu_\delta$ are both equal to ν_n, and the coefficient of the smallest power of y_κ namely of $y_\kappa^{\nu_n}$ in (22) is

$$2\begin{vmatrix} y_{\kappa+1}^{\nu_\kappa} \cdots y_n^{\nu_\kappa} \\ \cdots\cdots\cdots\cdots \\ y_{\kappa+1}^{\nu_{n-1}} \cdots y_n^{\nu_{n-1}} \end{vmatrix}^2$$

which is also not identically equal to zero. Thus we see that (21), equally as (20), can not be identically zero.

Since the coefficients of λ and λ^2 of (19) are not identically zero, there must be, in general, a value of λ, not equal to zero for which (19) will vanish. For this value of λ, we see that, in the expansion of (18), which is not identically zero, the coefficients of x^{ν_1}, x^{ν_2}, ..., x^{ν_n} are all zero. This contradicts the fact that S_{ν_1}, S_{ν_3}, ..., S_{ν_n} form a fundamental system.

Therefore we see that the sequence M must be additive. And hence our condition is necessary for all values of n.

Remark. It is naturally supposed that every fundamental system should be normal. But I have no proof for it.